Evaluation

EEZQUQ

Multiple choice questions

1. An dye has negative charge.
   a. Basic
   b. Acidic
   c. Neutral d. None
2. _____ stain is incorporated with Eosin in Lactose agar to distinguish typical *Escherichia coli* in contaminated water.
   a. Crystal violet b. Acid fuchsin
   c. Methylene blue d. Safranin
3. Which of the following is not an anionic dye?
   a. Safranin b. Eosin
   c. Rose Bengal d. Acid fuchsin
4. Christian Gram discovered a staining technique to differentiate the bacteria of similar morphology in the year.
   a. 1857 b. 1880 c. 1884 d. 1881
5. Which of the following is used for negative staining of microbial cells?
   a. Nigrosin and Acid fuchsin
   b. Rose Bengal and malachite green
   c. Safranin and Eosin
   d. Nigrosin and Indian Ink
6. _____ is used as a mordant in Gram staining techniques.
   a. Iodine
   b. Crystal violet
   c. Methylene blue
   d. Safranin
7. Which of the following pairs is mismatched?
   a. Capsule-negative stain
   b. Cell arrangement-simple stain
   c. Cell size-albert stain
   d. Gram stain-bacterial identification
8. The order of reagents in the gram staining reactions are:
   a. Safranin, alcohol, methylene blue, iodine
   b. Crystal violet, iodine, alcohol, safranin
   c. Methylene blue, alcohol, iodine, safranin
   d. Crystal violet, alcohol, iodine, safranin
9. The Schaeffer-Fulton endospore staining usually shows
   a. Spore green within pink cells
   b. Spores pink within green cells
   c. Colourless spores within pink cells
   d. Colourless spores within green cells

Answer the following

1. Define stain.
2. Give examples for basic stain.
3. Why heat fixation is important?
4. What are endospores?
5. Distinguish between a dye and a stain.
6. List out few gram positive bacteria.
7. What is the purpose of a counterstain/ decolorizer in the gram stain?
8. Fill in the following table regarding the gram stain.

	Appearance after this step of gram staining	
Steps	Gram positive cells	Gram negative cells
Crystal violet		
Iodine		
Alcohol		
Safranin		

9. What is meant by negative staining?
10. What are the uses of staining?
11. Differentiate simple and differential stain.
12. What are acidic stains? Give examples.
13. Why do basic dyes stain bacterial cells? Why won't acidic dyes stain bacterial cells?

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